

2026 Argonia Cup Rules

SUMMARY

The goal for the 2026 Argonia Cup is to design and launch an efficient and optimized two-stage rocket containing any number of golf balls as a payload in the sustainer to the highest altitude possible. The sustainer must reach a minimum of 9,000' AGL and not exceed 40,000' AGL. Both stages of the rocket must recover safely and in re-flyable condition.

RULES

1. There must be at least one (1) TRA or NAR certified Level 2 member per team. This team member must be present at the competition and will be considered the flyer of record of the rocket. Teams must be made up of two or more students. All team members must currently be enrolled at the competing university or college. Team members accessing the high-power pad area must be members of the Tripoli Rocketry Association or the National Association of Rocketry (NAR). Multiple teams from the same university or college are permitted.
2. The maximum total installed impulse for this competition will be 5,120.0 Newton Seconds. All motors must be commercially certified motors. The certified impulse for motors flown will come from the Combined CAR/NAR/TRA Certified Rocket Motors List found [here](#). Spark-emitting motors (Skidmark-type motors) are prohibited.
3. A commercially manufactured and tested altitude recording altimeter with onboard data storage shall be used for altitude determination. If two or more altimeters are used, the average apogee height of each altimeter will be used for the determination of the rocket's apogee.
4. Sustainer apogee will be recorded from altimeter beeps (or a witnessed data download) at the judging table upon return from successful recovery. It is required that altimeters be programmed to report apogee in feet.
5. Ignition of the sustainer motor must utilize commercially manufactured electronics capable of inhibiting sustainer ignition if the flight is not vertical. See the sustainer Motor Inhibit Logic section below for details.
6. All motor igniters shall be installed in rocket motors at a safe location out on the range. Any team found installing ignitors in the team camp/spectator area will be disqualified.
7. All flights will be angled away from the flight line regardless of wind direction.
8. All rockets must have a minimum of a 5:1 thrust-to-weight ratio at liftoff.
9. All rocket second stages must have a minimum of a 2:1 thrust-to-weight ratio at sustainer motor ignition.
10. Launch configuration flight stability shall be achieved by maintaining a minimum CP/CG static margin of no less than 1 body caliber during flight.
11. All launch vehicle components must be presented to the judging table as recovered. Rockets must be recovered in accordance with the Tripoli Unified Safety Code (TUSC). Rockets must be recovered in a "re-flyable condition" after flight without excessive damage, or they will be considered disqualified. Excessive damage is defined as:
 - a. Damage to the rocket that would prevent flying it again safely.
 - b. A zipper of more than a quarter caliber.
 - c. Non-cosmetic damage to fillets or fins, such as a fin tab that is no longer securely bonded to either the motor mount tube or body tube.
 - d. A failed recovery system that allows the rocket to descend faster than designed, even if there is no damage to the rocket.

SUSTAINER MOTOR INHIBIT LOGIC

Sustainer motor ignition will be inhibited in the event the flight is not vertical. This can be done in one of two ways.

1. Use flight computers capable of inhibiting sustainer motor ignition based on tilt angle.
 - ❖ Examples include the Altus Metrum TeleMega and EasyMega, Featherweight Blue Raven, and Silicdyne Fluctus. Other devices also exist that sense tilt angle and use this information as a check prior to igniting the upper stage.
- The following logics should all be true before firing the upper stage:
 - Tilt angle less than 20 degrees.
 - Velocity greater than 200 ft/sec
2. Time/altitude lockout
 - Devices without tilt measurements are allowed, but should adhere to the following guidelines:
 - Velocity greater than 200 ft/sec
 - Time since liftoff less than (intended staging time + 2 sec)
 - Altitude greater than (75% of the intended staging altitude)

NOTE: The sustainer recovery system should be sized to allow safe descent of the sustainer in the event that the sustainer motor does not ignite. Keep in mind that unburned propellant will make the rocket heavier. A descent rate of no more than 35 ft/sec will be allowed under the main parachute with an unburned rocket motor.

MAJOR DEADLINES & TIMELINE OF COMPETITION WEEKEND

- Video submission deadline: February 20, 2026
- Flight simulation deadline: February 20, 2026
- Vehicle frequency deconfliction release: March 14, 2026
- Competition Weekend: March 21-22, 2026

VIDEO

Each team will be required to produce a video (eight (8) minutes or longer, hosted on YouTube) introducing the team and providing a clear, detailed overview of their vehicle and design methodology. Additional emphasis should be placed on the safety and design of the rocket staging layout. This video must be publicly available until the competition date and submitted to the launch organizers on or before February 20, 2026, [here](#). Failure to meet these requirements by the required deadline will be grounds for disqualification.

FLIGHT SIMULATIONS

Team information, detailed rocket description, vehicle transmitting frequencies (telemetry and radio beacons), flight simulations, and sustainer logic information must be submitted to the contest committee by February 20, 2026, at kloudbustersargoniacup@gmail.com. The information submission form can be found [here](#). Failure to submit this information by the required deadline will be grounds for disqualification.

FREQUENCY DECONFLICTION

After communicating with each team's intended frequency usage, the Argonia Cup committee will work to minimize conflicts between teams. This information will be released to the teams at least one week prior to the competition weekend.

COMPETITION LAUNCH OPERATIONS

1. TRA Unified Safety Code will be followed for all launch activities. Any team member accessing the launch pad area must be a member of TRA or the National Association of Rocketry (NAR).
2. The launch organizers will provide all launch pads, launch rails, and the launch control system. All rockets must utilize 1.5" x 1.5" (commonly known as 1515) rails. The rails provided will be 12' in length. A minimum of two (2) rail guides must be used on the booster.
3. All rockets will undergo a rocket safety inspection before the teams will be cleared to fly their projects. Any safety of flight issues noted in this inspection will be resolved before flight. These safety inspectors have the final say regarding any project's suitability for flight.
4. All rockets must be flown between 9:00 AM and 5:00 PM on Saturday, March 21st and 9:00 and the last declared bank (approximately 4:00 pm) on Sunday, March 22nd.
5. All rockets must be presented to the scoring table no later than 45 minutes after the completion of the last competition flight on Sunday, March 22nd, 2026. Flights returned after this time will be disqualified.
6. Teams will only be allowed three launch attempts to light both stages and receive a qualifying score.
7. Teams will only be allowed to rack a single vehicle for a launch attempt at one time.
8. Teams will be given 30-minute windows (starting when the first team arrives at the pad bank) to rack and prepare their rockets for flight. Teams that are not ready to launch at the close of their window will be required to stand down to allow fully prepared teams a chance to fly. They will not lose their position in line for a pad.

SCORING

1. The flight score for this competition will be as follows:

$$Score = Apogee_{sustainer} * \left(1 + \frac{-I_T^2 + 5120I_T}{6540806}\right) * \left(2 - e^{-\frac{n}{6}}\right)$$

Where:

" $Apogee_{sustainer}$ " is the apogee of the second stage in feet.

" n " is the number of golf balls contained in the sustainer of the rocket.

" I_T " is the combined total impulse of the motors in the rocket in Newton-seconds.

" e " is the mathematical constant known as Euler's number.

2. Teams must supply all golf balls they intend to use in the competition. All golf balls must conform to USGA specifications and cannot be modified in any way.
3. The minimum number of golf balls flown is one.
4. Competition golf balls may only be loaded under the supervision of the contest-appointed safety monitor.
5. Only golf balls flown and recovered in the sustainer will be considered for scoring.
6. Any altitude less than 9,000' AGL or greater than 40,000' AGL will be disqualifying.